

Zachary Porat

Department of Mathematics and Statistics
Bucknell University
380 Olin Science Building
Lewisburg, PA 17837

✉ z.porat@bucknell.edu
🖱 <https://zporat.github.io>
🌐 U.S. Citizen
📅 Last Updated October 1, 2026

EMPLOYMENT

Bucknell University <i>Visiting Assistant Professor</i>	Lewisburg, PA 2026 – Present
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EDUCATION

Wesleyan University <i>Ph.D. in Mathematics</i>	Middletown, CT 2020 – 2026
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- Thesis: “On the Cuspidal Cohomology of Congruence Subgroups of $SL(3, \mathbb{Z})$ ” [[Link](#)]
- Advisor: David Pollack

Union College <i>B.S. in Mathematics with Honors, summa cum laude</i>	Schenectady, NY 2016 – 2020
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- Honors Thesis: “Classification of Torsion Subgroups for Mordell Curves” [[Link](#)]
- Physics Minor, Phi Beta Kappa, Scholars Program

RESEARCH INTERESTS

Broad: number theory, arithmetic geometry

Specific: automorphic forms, cohomology of arithmetic groups, arithmetic statistics, algebraic curves, computational aspects of number theory and arithmetic geometry

PUBLICATIONS AND PREPRINTS

- [6] **On the Cuspidal Cohomology of Iwahori Congruence Subgroups of $SL(3, \mathbb{Z})$** , submitted (2026). [[Code](#)]

🔗 [arXiv:2609.01776](https://arxiv.org/abs/2609.01776)

- [5] **Distinguishing Elliptic Curves Modulo p and Identifying Images of Product Representations** (with Jessica Bennett, Jeffrey Hatley, Sasha Kononova, Vincent Macri, and Naina Praveen), submitted (2026). [[Code](#)]

🔗 [arXiv:2608.10880](https://arxiv.org/abs/2608.10880)

- [4] **Heuristics for (Ir)reducibility of p-Rank Strata of the Moduli Space of Hyperelliptic Curves** (with Thomas Bouchet, Erik Davis, Steven R. Groen, and Benjamin York), *Experimental Mathematics* (2026). [Code]
 DOI: [10.1080/10586458.2026.2671126](https://doi.org/10.1080/10586458.2026.2671126)
- [3] **Computations Directly on the Cuspidal Cohomology of Congruence Subgroups of $SL(3, \mathbb{Z})$** , *Mathematics of Computation* (2025). [Code]
 DOI: [10.1090/mcom/4155](https://doi.org/10.1090/mcom/4155)
- [2] **Family Sizes for Complete Multipartite Graphs** (with Danielle Gregg, Thomas Mattman, and George Todd), *Involve, a Journal of Mathematics*, Vol. 15, No. 4 (2022), 669-686.
 [MR4536581](https://www.ams.org/mathscinet/MR4536581), DOI: [10.2140/involve.2022.15.669](https://doi.org/10.2140/involve.2022.15.669)
- [1] **PIXE Analysis of Synthetic Turf** (with Sajju Chalise, Morgan Clark, Skye Conlan, Scott LaBrake, and Michael Vineyard), *Environment and Ecology Research*, Vol. 6, No. 1 (2018), 60-65.
 DOI: [10.13189/eer.2018.060105](https://doi.org/10.13189/eer.2018.060105)

TEACHING EXPERIENCE

Bucknell University

Instructor of Record

Fall 2026 – Present

- MATH 245 - Linear Algebra (Fall 2026)
- MATH 201 - Calculus I (Fall 2026 $\times 2$)

Wesleyan University

Instructor of Record

Fall 2022 – Spring 2026

- MATH 500 - Graduate Pedagogy (Fall 2025)
- MATH 121 - Calculus I (Fall 2024)
- MATH 132 - Elementary Statistics (Fall 2022)

Graduate Teaching Assistant

Fall 2021 – Spring 2026

- MATH 221 - Vectors and Matrices (Spring 2026)
- MATH 261 - Introduction to Abstract Algebra (Fall 2023, Fall 2025)
- MATH 225 - Introduction to Real Analysis (Spring 2022, Spring 2023, Spring 2025)
- MATH 262 - Advanced Topics in Abstract Algebra (Spring 2024)
- MATH 228 - Discrete Mathematics (Fall 2021)

Directed Reading Program Mentor

Spring 2023 – Spring 2025

- William Allen, *Elliptic Curve Cryptography: Data Encryption through Math* (Spring 2025)
- William Dollhopf, *Introduction to Cryptography and the ElGamal Key Cryptosystem* (Fall 2023)

- Emily Bennett, *Understanding PKC: The Diffie-Hellman Key Exchange* (Spring 2023)
- Stefan Hesseling, *RSA and Integer Factorization* (Spring 2023)

Math Workshop Tutor

Fall 2020 – Spring 2021

- Covers the entire selection of undergraduate math courses at Wesleyan

Union College

Course Assistant

Fall 2017 – Winter 2020

- MTH 340 - Linear Algebra (Winter 2020)
- MTH 110 - Calculus I (Fall 2017, Fall 2018, Fall 2019)
- MTH 112 - Calculus II (Winter 2018, Winter 2019)
- MTH 115 - Differential Vector Calculus and Matrix Theory (Spring 2018)

Calculus Help Center Tutor

Fall 2019 – Winter 2020

- Covers Calculus I, Calculus II, and Multivariable Calculus with Linear Algebra

OUTREACH

Project Leader (with Jeffrey Hatley)

June 22 – July 3, 2026

Rethinking Number Theory [[Link](#)] [See [Paper #5](#)]

Virtual

Project Assistant: Iwasawa Theory and Applications

May 5-9, 2025

Arithmetic Geometry at UNT [[Link](#)]

Denton, TX

Mini-Course Lecturer: Mathematical Cryptography [[Notes](#)]

March 29-30, 2025

Student Workshop in Mathematics [[Link](#)]

Middletown, CT

INVITED TALKS

Joint Mathematics Meetings

January 13, 2027

Special Session: Explicit Methods in Number Theory

Chicago, IL

Lafayette College Math Colloquium

October 29, 2026

Lafayette College

Easton, PA

Bucknell University Math Colloquium

April 3, 2026

Bucknell University

Lewisburg, PA

CSUB Math Seminar

March 27, 2026

California State University, Bakersfield

Bakersfield, CA

UIC Number Theory Seminar

March 13, 2026

University of Illinois Chicago

Chicago, IL

Five College Number Theory Seminar

February 10, 2026

Amherst College

Amherst, MA

Wesleyan Algebra and Number Theory Seminar <i>Wesleyan University</i>	February 6, 2026 Middletown, CT
UConn Algebra Seminar <i>University of Connecticut</i>	December 3, 2025 Storrs, CT
AMS Fall Eastern Sectional Meeting <i>Special Session: Number Theory Meets Algebraic Geometry</i>	October 25, 2025 Virtual
AMS Fall Southeastern Sectional Meeting <i>Special Session: Modular Forms in Combinatorics and Number Theory</i>	October 4, 2025 New Orleans, LA
AWM Research Symposium <i>Special Session: Curves and Abelian Varieties in Characteristic p</i>	May 16, 2025 Madison, WI
AMS New England Graduate Student Conference <i>Brown University</i>	April 19, 2025 Providence, RI
Manhattan College Math Seminar <i>Manhattan College</i>	November 15, 2023 Bronx, NY
Union College Math Seminar <i>Union College</i>	January 12, 2023 Schenectady, NY

CONTRIBUTED TALKS

33èmes Journées Arithmétiques <i>Université du Luxembourg</i>	July 4, 2025 Belval Esch-sur-Alzette, LUX
37th Automorphic Forms Workshop <i>University of North Texas</i>	May 4, 2025 Denton, TX
Conférence de Théorie des Nombres Québec-Maine <i>Université Laval</i>	October 6, 2024 Québec City, CAN
36th Automorphic Forms Workshop <i>Oklahoma State University</i>	May 24, 2024 Stillwater, OK
Graduate Student Seminar <i>Wesleyan University</i>	March 9, 2023 Middletown, CT
Graduate Student Seminar <i>Wesleyan University</i>	October 14, 2021 Middletown, CT

SELECT WORKSHOPS AND CONFERENCES

Rethinking Number Theory[†] <i>Distinguishing Elliptic Curves and Understanding Product Representations</i>	June 22 – July 3, 2026 Virtual
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[†]Denotes that I served as a project leader at this workshop. See [Outreach](#) for additional details. See also [Paper #5](#).

Arizona Winter School <i>Computational Aspects of Arithmetic Geometry and Cryptography</i>	March 7-11, 2026 Tucson, AZ
LMFDB, Computation, and Number Theory (LuCaNT) 2025 <i>Institute for Computational and Experimental Research in Mathematics</i>	July 7-11, 2025 Providence, RI
Arithmetic Geometry at UNT[†] <i>University of North Texas</i>	May 5-9, 2025 Denton, TX
Algorithmic Number Theory Symposium (ANTS) XVI <i>Massachusetts Institute of Technology</i>	July 15-19, 2024 Cambridge, MA
Arizona Winter School <i>Abelian Varieties</i>	March 2-6, 2024 Tucson, AZ

PROFESSIONAL SERVICE

Reviewer for zbMATH Open <i>zbMath Open</i>	Spring 2025 – Present Virtual
Workshop Organizer (with Lydia Ahlstrom and Arianna Zikos) <i>Student Workshop in Mathematics</i> [Link]	March 2025 Middletown, CT

DEPARTMENTAL SERVICE

DRP Organizer (with Arianna Zikos) <i>Wesleyan Directed Reading Program</i> [Link]	Spring 2023 – Spring 2025 Middletown, CT
President <i>Wesleyan AMS Graduate Student Chapter</i>	Fall 2023 – Spring 2025 Middletown, CT
Seminar Organizer (with Justin Bryant) <i>Wesleyan Graduate Student Seminar</i>	Spring 2022 – Fall 2023 Middletown, CT
Vice President <i>Wesleyan AMS Graduate Student Chapter</i>	Fall 2021 – Spring 2023 Middletown, CT

HONORS AND AWARDS

Martin Terry Resch Prize [Link] <i>Union College, Department of Mathematics</i>	May 2020 Schenectady, NY
Phi Beta Kappa Membership <i>Union College Chapter</i>	May 2020 Schenectady, NY
Pi Mu Epsilon Membership <i>Union College Chapter</i>	April 2019 Schenectady, NY

[†]Denotes that I served as a project assistant at this workshop. See [Outreach](#) for additional details.

GRANTS

AMS Sectional Travel Grant*AMS Fall Southeastern Sectional Meeting*October 2025
New Orleans, LA**AWM Research Symposium Travel Grant***AWM Research Symposium*May 2025
Madison, WI

SKILLS

Computer: Magma, SageMath, Python, Linux, $\text{\LaTeX}$ **Language:** English (native), French (proficient)